TSM 099

Homework 4

Simplifying, multiplying, and dividing fractions

Topics 18-20 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers. State every restriction from the original expression, including any value that makes a divisor zero. Cancel common factors, not terms.

Numerical structure

1. Compute and reduce.

$$\frac{15}{28} \cdot \frac{14}{25}$$

2. Compute and reduce.

$$\frac{-7}{12} \div \frac{14}{9}$$

Simplifying rational expressions

3. Simplify and state original restrictions.

$$\frac{x^2 - 16}{x^2 + 7x + 12}$$

4. Simplify and state original restrictions.

$$\frac{3x^2 - 12x}{6x}$$

5. Simplify and state original restrictions.

$$\frac{x^2 - 6x + 9}{x^2 - 9}$$

Products and quotients

6. Multiply and simplify. State every original restriction.

$$\frac{x^2 - 4}{x^2 - 5x + 6} \cdot \frac{x - 3}{x + 2}$$

7. Multiply and simplify. State every original restriction.

$$\frac{x^2 - 9}{x^2 + x - 6} \cdot \frac{x - 2}{x^2 - 3x}$$

8. Divide and simplify. Include restrictions caused by the divisor.

$$\frac{x^2 - 1}{x + 3} \div \frac{x - 1}{x + 2}$$

9. Divide and simplify. Include restrictions caused by the divisor.

$$\frac{2x}{x-4} \div \frac{x^2}{x^2-16}$$

10. Explain why canceling x from (x + 2)/x is invalid. Rewrite the expression as a sum of two terms for nonzero x.

11. A simplified expression equals 1, but its original form was (x squared - 4)/[(x - 2)(x + 2)]. State its domain and explain why it is not the constant function 1 on all real numbers.

12. For Problem 8, check the original and simplified expressions at x = 2. Explain why the restriction x ≠ 1 remains even though a factor canceled.